

Samuel Chau

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Summary

Computer Science graduate and maintainer of Profilarr, an open-source configuration-as-code tool for self-hosted media servers with 3.5M+ downloads and 2,400 stars. Experienced across full-stack web development (SvelteKit, Deno, TypeScript), systems programming, and cybersecurity, with a track record of designing developer tools that scale.

Education

Bachelor • University of Adelaide, Computer Science, Major in Cybersecurity • GPA: 6.4/7.0 — Achieved 13 High Distinctions • Top Results: Databases & Ethical Data (96), Web & Database Computing (95), Algorithm & Data Structure Analysis (93), Software Engineering & Project (92)	Adelaide, SA Feb 2022 – Dec 2025
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Experience

Profilarr, Maintainer & Lead Developer Open-source tool that brings configuration-as-code to self-hosted media servers. • Designed a CRDT-inspired, event-sourced configuration system — config is stored as operations rather than final state, so upstream curators and local users can edit independently and merge without conflict; value guards catch same-field edits before they silently overwrite. • Architected and built a 100K+ line SvelteKit/Deno/TypeScript/SQLite app solo, including a Terraform-style sync engine (idempotent, retry-safe, with drift detection), a crash-recoverable job queue, and a fully type-safe data layer generated directly from the database schema and queried through Kysely. • Built a reusable component library that leverages semantic CSS tokens to create a consistently styled, themable, and polished design, enabling rapid UI development. Enforced using custom Svelte-AST lint rules that ban raw HTML and hardcoded styles. • Designed a defense-in-depth security model that keeps raw secrets off the web surface entirely — authenticated or not: clients receive presence flags rather than values, the app's own API key is hashed like a password (shown once, never retrievable), and backup downloads are credential-stripped before reaching the browser, leaving the filesystem as the only trust boundary for full-fidelity data. Layered with independent OWASP Top 10 protections: sanitized markdown, path-traversal and reverse-proxy-aware CSRF handling, and persistent rate limiting. • Built a CI/CD release pipeline running build, lint, type checks, unit + integration tests against the production binary, Semgrep SAST, and Playwright E2E auth flows on every change. • Grew to 3.5M+ downloads and 2,400 GitHub stars ran all support, beta coordination, and contribution guidelines for a 2,000+ member community.	Apr 2023 – present 3 years 3 months
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Projects

Synopsis AI Research Paper Video Platform • Built full-stack platform using Grobid for paper parsing, LLMs for summarization/image generation, and FFMpeg for video synthesis. • Pitched at Tech eChallenge startup competition; ranked Top 5 out of 50+ teams.	React, Express, LLMs, FFMpeg
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OMesh

Python, Docker, WebSockets

Secure Messaging Platform

- Built end-to-end encrypted messaging platform using low-level socket programming and custom protocol design.
- Implemented RSA/AES-GCM cryptographic pipeline for secure key exchange and message encryption.
- Deployed multi-server architecture with Docker Compose; real-time communication via WebSockets.

Rogue

C++, SFML

Procedural Roguelike Game

- Built turn-based roguelike with procedural map generation and entity component system architecture.
- Implemented design patterns (Factory, Observer, State) for extensible game object behavior.

Skills

Languages: TypeScript, Python, C++, C#, SQL, HTML/CSS

Technologies: SvelteKit, Deno, Docker, Git, Linux, SQLite, Kysely, Discord.js, FFmpeg

Practices: GitOps, Event Sourcing, API Design, Open Source Development

Interests: Self-Hosting, Homelabbing, Media Archival, Basketball, Technical Writing